MediScope

Transforming Raw Patient Datasets into Actionable Healthcare Insights

Team G-17 | Newton School of Technology • Mentor: Archit Raj • April 2026

Context & Problem Statement

The Challenge

Hospital systems lack visibility into:

- Which patient sub-populations drive extended ICU stays
- Which lab departments generate disproportionate volume vs. diagnostic yield

Without this data-driven insight, leadership cannot:
✓ Allocate beds & staff proactively
✓ Distinguish signal from noise in test results

Three Core Findings

1 Emergency Dominance

91% emergency cases, 36% longer stays

2 Gender Disparity

Female patients flag 4–5 pts higher ($p < 0.0001$)

3 Hematology Drain

Highest volume + highest abnormal rate

Data Engineering & ETL Pipeline

SOURCE		TRANSFORM		LOAD
MIMIC-III ICU Records	→	Python/Pandas Data Cleaning	→	Tableau Dashboards

Final Dataset:	76,069 laboratory event rows
Unique Patients:	100 ICU patients
Total Admissions:	129 admission records
Key Variables:	Subject ID, Admission ID, Test Type, Abnormal Flag, LOS, Admission Type, Gender
Data Quality:	99.99% retained after cleaning (5 rows with unresolvable nulls removed)

KPI & Metrics Framework

76,069

Total Lab Tests

Scale of testing activity

Total Records

76,069

38.79%

Abnormal Rate

vs. 10-15% in outpatient

Anbormal Rate

0.3879

22.5 days

Average LOS

Length of stay baseline

Average Stay

22.51

760

Test Per Patient

Average Test

Test per patient

760.7

Key Insights from Exploratory Analysis

1. Emergency admission dominance: 91% of volume, 36% longer LOS than elective (28.8 vs. 21.2 days)

2. Gender disparity: Female patients consistently flag higher abnormal rates across all admission types ($p < 0.0001$)

3. Hematology bottleneck: Highest test volume (33,567 tests) AND highest abnormal rate (44.5%)

4. Volume-rate divergence: Test volume spikes peak at 9,200/month but abnormal rate stays flat at 39% — signature of systematic over-testing

Advanced Analysis & Statistical Validation

Independent T-Test

Emergency vs. Elective LOS

Δ = 7.6 days (+36%)

Result:

$p < 0.0001$

Chi-Square Test

Gender vs. Abnormal Flag

Female > Male (all types)

Result:

$p < 0.0001$

K-Means Clustering

Patient Risk Segmentation

3 clusters identified (n=60,26,14)

Result:

Cluster 2: Critical outliers

All findings are statistically significant and reproducible

Dashboard Architecture

3-Tier Executive Decision-Support Framework

TIER 1	**Macro KPIs** Total tests, abnormal rates, patient load, LOS baseline	*C-suite (30-second overview)*
TIER 2	**Demographic & Operational Drivers** Gender disparities, LOS by admission type, risk segments	*Operations leaders (5-min deep dive)*
TIER 3	**Clinical Deep Dive** Dept. quadrant analysis, volume spikes, test justify signals	*Department heads (bottleneck isolation)*

Dashboard published at: public.tableau.com/app/profile/sambhav.kumar2781/viz/dv_p7/Dashboard1

Actionable Recommendations (1 of 2)

1. Optimize Emergency Triage Workflows

Fast-track discharge protocols + daily LOS review rounds (18-25 day window)

Est. Impact: 10% LOS reduction = 264 recovered bed-days per 100-patient cycle

2. Introduce Gender-Specific Laboratory Reference Ranges

Audit D_LABITEMS, recalibrate female-specific thresholds, 30-day parallel validation

Est. Impact: 5–10% reduction in false-positive alerts = ~2,083 fewer physician review events

Actionable Recommendations (2 of 2)

3. Implement Hematology Repeat-Testing Hard Stops in EHR

Mandatory clinical justification for 24-hr repeat panels + peer benchmarking dashboards

Est. Impact: 15% volume reduction + reagent savings + lower alert load on nursing

4. Deploy Prospective LOS Prediction Model for Critical Outliers

K-Means Cluster 2 (14 critical patients) → supervised model → admission-day flagging

Est. Impact: Early identification reduces reactive escalation (2–3x cost premium). Requires 1K+ patient dataset.

Quantified Impact Summary

Emergency LOS Optimization

264 bed-days recovered per 100-patient cycle

Gender Reference Ranges

~2,083 fewer false-positive alerts per cycle

Hematology Hard Stops

15% volume reduction + significant reagent savings

Predictive LOS Model

2–3x cost savings (post dataset expansion)

Limitations & Future Scope

Current Limitations

- Sample size (100 patients) sufficient for hypothesis testing but limiting for ML model training
- No direct cost data — all financial estimates use published benchmarks, not actual billing data
- Single institution, specific time window — regional, seasonal, and case-mix variations not captured

Phase 2 Roadmap

→ Dataset expansion to 1,000+ patients (full MIMIC-III cohort) for robust ML training

→ Integration of hospital billing & cost data for precise Cost-Benefit Analyses

→ Real-time EHR dashboard integration for prospective operational monitoring

Team & Contributions

Satyam Kumar	Project Lead & Data Engineer	*ETL pipeline, architecture, integration*
Sambhav Kumar	Visualization Specialist	*Tableau dashboard design & dashboards*
Shagun Singh	Statistical Analyst	*T-tests, Chi-Square, K-Means clustering*
Yash	Documentation Lead	*Report generation & documentation*
Harsh	Research & QA	*Project review & presentation*

Faculty Mentor: Archit Raj | Newton School of Technology | April 2026

GitHub: github.com/Satyamkumar2610/SectionD_G-17_MediScope